BIDERIVATIONS FROM A POLYNOMIAL RING ARE DETERMINED BY THEIR VALUES ON THE VARIABLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a two-line construction with the displayed monomial formula; uniqueness follows because the values on monomials are forced by the derivation rules.

Editorial note. The counted spans carry no external references, so no navigation stubs are needed beyond the ones implied by the statement wording.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The definition of an A-biderivation from a polynomial algebra. used at proof lines 2648-2650. Fixes the object whose existence and uniqueness are asserted.
- The formula for a derivation on monomials. used at proof lines 2653-2659. Provides the explicit construction.
- A-linearity and the bimodule structure. used at proof lines 2659-2659. Extends the formula to all of P and yields the biderivation.

Lemma 0.2. *Let $P = A[S]$ be a polynomial ring over A. Let M be a (P, P)-bimodule over A. Given $m_s \in M$ for $s \in S$, there exists a unique A-biderivation $\lambda : P \to M$ mapping s to m_s for $s \in S$.*

Proof. We set

$$\lambda(s_1 \ldots s_t) = \sum s_1 \ldots s_{i-1} m_{s_i} s_{i+1} \ldots s_t$$

in M. Extending by A-linearity we obtain a biderivation. □

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